

FILE



California Regional Water Quality Control Board

Central Valley Region

Robert Schneider, Chair



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Secretary for
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Governor

23 February 2004

Mr. Felix Ortiz
Director of Public Works
P.O. Box 237
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REVISED MONITORING AND REPORTING PROGRAM NO. R5-2002-0063, CITY OF EXETER, WASTEWATER TREATMENT FACILITY, TULARE COUNTY

Enclosed is a copy of Revised Monitoring and Reporting Program Order No. R5-2002-0063. Based on your 17 December 2003 letter and subsequent discussion between Mr. Stephen Klein of Regional Board staff and your consultant, Carollo Engineers, you concur with the revisions. You are now required to immediately implement the revised Monitoring and Reporting Program.

If you have any questions regarding this matter, please contact Stephen Klein at (559) 445-5558.

DOUGLAS K. PATTESON
Senior Engineer
RCE No. 55985

Enclosure: Revised Monitoring and Reporting Program No. R5-2002-0063

cc: Mr. Roy Chace, City Administrator
Mr. David Stringfield, Carollo Engineers, Fresno
Ms. Kristie D. Sandoval, Carollo Engineers, Fresno

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California Environmental Protection Agency



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CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

REVISED MONITORING AND REPORTING PROGRAM NO. R5-2002-0063
FOR
CITY OF EXETER
WASTEWATER TREATMENT FACILITY
- TULARE COUNTY

This Monitoring and Reporting Program (MRP) is required pursuant to California Water Code section 13267. The Discharger shall not implement any changes to this MRP unless and until the Regional Board adopts or the Executive Officer issues a revised MRP. Influent and effluent sample station locations are depicted on Attachment B. Changes to these sample locations shall be established with concurrence of Regional Board's staff, and a description of the revised stations shall be submitted to the Regional Board and, following approval of the Executive Officer, attached by the Discharger to its copy of this Order. All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each sample shall be recorded on the sample chain of custody form. All analyses shall be performed in accordance with Standard Provisions, Provisions for Monitoring.

INFLUENT MONITORING

The Discharger shall collect influent samples at the headworks of the treatment facility prior to any treatment of waste. Time of a grab sample shall be recorded. Influent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Maximum Daily Flow	mgd	Continuous	Continuous
Average Daily Flow	mgd	Computed	Daily ¹
Monthly Average Flow	mgd	Computed	Monthly
Settleable Solids	mL/L	Grab	2/week ²
pH	pH units	Grab	2/week ²
BOD ₅ ³	mg/L	8-hr Composite ⁴	2/week ²
Monthly Average BOD ₅	mg/L	Calculated	Monthly
TSS ⁵	mg/L	8-hr Composite ⁴	2/week ²
Monthly Average TSS	mg/L	Calculated	Monthly

¹ Sample frequencies referenced hereafter in this program as daily shall not include weekends or holidays.

² On nonconsecutive days

³ Five-day, 20°C biochemical oxygen demand

⁴ 8-hour composite sampling as referred to in this program shall be flow-proportioned.

⁵ Total Suspended Solids

EFFLUENT MONITORING

The Discharger shall collect effluent samples at a point in the system following treatment and before discharge to the ponds (or Use Area, if applicable). Effluent samples shall be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency¹</u>
Settleable Solids	mL/L	Grab	Daily
pH	pH Units	Grab	Daily
Total Coliform Bacteria	MPN/100 ml	Grab	Daily ^{2,3}
BOD ₅			
Concentration	mg/L	8-hr Composite	Weekly
Monthly Average	mg/L	Calculated	Monthly
Percent Removal	%	Calculated	Monthly
TSS			
Concentration	mg/L	8-hr Composite	Weekly
Monthly Average	mg/L	Calculated	Monthly
Percent Removal	%	Calculated	Monthly
Total Dissolved Solids (TDS) ⁴	mg/L	8-hr Composite	2/Month ^{5,6}
EC ⁷	µmhos/cm	8-hr Composite	2/Week
Nitrate (as N)	mg/L	8-hr Composite	2/Month ⁶
Total Kjeldahl Nitrogen (TKN)	mg/L	8-hr Composite	2/Month ⁶
Total Nitrogen	mg/L	Calculated	2/Month ⁶
Ammonia	mg/L	8-hr Composite	2/Month ⁶
General Minerals ⁸	mg/L	Grab	Semiannually ⁹
Title 22 constituents ¹⁰	varies	8-hr Composite or Grab, whichever is appropriate	Tri-annually ¹¹

¹ If results of monitoring a pollutant appear to violate discharge specifications, but monitoring frequency is not sufficient to validate violation (e.g., the monthly mean for BOD₅), or indicate a violation and potential upset of the treatment process, the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.

² If required by Discharge Specification B.18

³ Daily when Discharging to the Use Area

⁴ TDS referenced hereafter in this program shall be determined using Environmental Protection Agency (EPA) Method No. 160.1 for combined organic and inorganic TDS and EPA Method No. 160.4 for inorganic TDS or equivalent analytical procedures specified in 40 Code of Federal Regulations (CFR) Part 136.

⁵ After six months, may be reduced to semiannually (April and October).

⁶ In nonconsecutive weeks coincident with influent EC sampling.

⁷ Conductivity at 25°C

⁸ General Minerals as referred to in this program shall include the constituents in the General Minerals Analyte List presented below.

⁹ April and October

¹⁰ Title 22 constituents, as used in this program, shall refer to constituents identified in the technical report submitted pursuant to Provision G.11.

¹¹ To coincide with source water testing performed by the City of Exeter for the Department of Health Services.

General Minerals Analyte List

Alkalinity (as CaCO ₃)	Carbonate (as CaCO ₃)	Manganese
Aluminum	Chloride	Phosphate
Bicarbonate (as CaCO ₃)	Hardness (as CaCO ₃)	Potassium
Boron	Iron	Sodium
Calcium	Magnesium	Sulfate

General Minerals Sample Collection and Preservation: With the exception of effluent samples, samples placed in an acid-preserved bottle must first be filtered through a 0.45 µm nominal pore size filter. If field filtering is not feasible, samples shall be collected in unpreserved containers and submitted to the laboratory within 24-hours with a request (on the chain-of-custody form) to immediately filter then preserve the sample.

POND MONITORING

Ponds shall be sampled systematically for the parameters specified below. Freeboard shall be monitored on all ponds in use to the nearest one tenth of a foot. Pond monitoring shall include at least the following:

<u>Constituent/Parameter</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ¹
Dissolved Oxygen (DO)	mg/L	Grab ²	As Required ³
Freeboard	feet ⁴	Observation	Weekly

¹ If results of monitoring appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation or indicate a violation and potential upset of the treatment process (e.g., less than minimum dissolved oxygen concentration), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.

² Samples shall be collected at a depth of one foot from each pond in use, opposite the inlet, and analyzed for DO. Samples shall be collected between 0700 and 0900 hours. If DO results for any pond in use indicate noncompliance with the effluent limit, the Discharger shall implement corrective measures as specified in the operation and maintenance manual and monitor said pond daily until its DO stabilizes above 1 mg/L.

³ If offensive odor detected by or brought to the attention of WWTF personnel, monitor affected pond(s) daily until dissolved oxygen > 1.0 mg/L. If DO results for any pond in use indicate noncompliance with Discharge Specification B.9, the Discharger shall implement corrective measures as specified in the O&M manual and monitor said pond daily until its DO stabilizes above 1.0 mg/L.

⁴ Freeboard shall be monitored to the nearest tenth of a foot.

In addition, the Discharger shall inspect the condition of ponds once per week and write visual observations in a bound logbook. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the ponds (e.g., dark sparkling green, dull green, yellow, gray, tan, brown, etc.). A summary of the entries made in the log during each month shall be submitted along with the monitoring report the following month. If the Discharger finds itself in violation of Discharge Specifications B.8, B.9, B.10, and B.11, the Discharger shall briefly explain the action taken or to be taken to correct the violation. The Discharger shall certify in each November monitoring report that it is in compliance with Discharge Specification B.16.

USE AREA MONITORING

The amounts of water and/or recycled water applied to the Use Area (in acre-feet) and amounts of chemical fertilizers (in pounds of nitrogen per acre) shall be measured and reported to the Regional Board quarterly in accordance with the following schedule:

<u>Monitoring Period</u>	<u>Data Due</u>
January – March	1 May
April – June	1 August
July – September	1 November
October - December	1 February

The Discharger shall utilize the form presented in Attachment G (or variation thereof subject to Regional Board staff approval) for reporting the Use Area monitoring data.

GROUNDWATER MONITORING

Prior to collecting samples and after measuring the water level, each monitoring well shall be adequately purged to remove water that has been standing within the well screen and casing that may not be chemically representative of formation water. Depending on the hydraulic conductivity of the geologic setting, the volume removed during purging is typically from 3 to 5 volumes of the standing water within the well casing and screen, or additionally the filter pack pore volume.

In the technical report submitted pursuant to Provision G.14 task f describing the results of the first sampling event performed pursuant to this program, the Discharger shall include a detailed description of the procedures and techniques for: (a) sample collection, including purging techniques, sampling equipment, and decontamination of sampling equipment; (b) sample preservation and shipment; (c) analytical procedures; and (d) chain of custody control. As it continues to monitor groundwater pursuant to this program, the Discharger shall report when it deviates from these procedures and techniques.

At least quarterly and concurrently with groundwater quality sampling, the Discharger shall measure the water level in each well as groundwater depth (in feet and hundredths) and as groundwater surface elevation (in feet and hundredths above mean sea level). The horizontal geodetic location for each monitoring well shall be provided where the point of beginning shall be described by the California State Plane Coordinate System, 1983 datum.

Samples shall be collected from approved monitoring wells and analyzed for the following constituents at the following frequency:

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CITY OF EXETER WWTF
TULARE COUNTY

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<u>Constituent/Parameter</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Depth to groundwater	To 0.01 foot hundredths	Measured	Quarterly ¹
Groundwater elevation	Above mean sea level, to 0.01 foot	Calculated	Quarterly ¹
⊗ pH	pH Units	Grab	Quarterly ¹
⊗ Total Organic Carbon	mg/L	Grab	Semiannually ²
Nitrogen compounds:			
⊗ Ammonia (as NH ₃ -N)	mg/L	Grab	Semiannually ²
⊗ Nitrate (as NO ₃ -N)	mg/L	Grab	Quarterly ¹
⊗ Total Kjeldahl Nitrogen (TKN)	mg/L	Grab	Semiannually ²
⊗ Total Organic Nitrogen (as N)	mg/L	Calculated	Semiannually ²
Salinity compounds/parameters:			
⊗ EC	μmhos/cm	Grab	Quarterly ¹
⊗ Chloride	mg/L	Grab	Quarterly ¹
⊗ Total dissolved solids	mg/L	Grab	Quarterly ¹
SAR ³	None	Calculated	Semiannually ²
General Minerals	mg/L	Grab	Semiannually ²

¹ January, April, July, and October

² April and October

³ Sodium adsorption ratio (SAR) = $\frac{Na}{\sqrt{\frac{Ca + Mg}{2}}}$, where Na, Cl, and Mg are in meq/L

Additionally, the Discharger shall include in the Provision G.14 task f technical report a technical description of proposed Data Analysis Methods for evaluating groundwater monitoring data (e.g., equivalent or similar to that described in Title 27, section 20415(e)(7-10)), consisting, at a minimum, methods to: (a) characterize natural background water quality of monitored constituents; (b) determine statistically significant differences between background and compliance wells for constituents that do not have water quality objectives or have background concentrations that exceed water quality objectives; and (c) select the minimum sample size required for the proposed data analysis approach and, if greater than that required by this program (i.e., quarterly), identification of when and how the additional samples will be collected during the one-year groundwater characterization period.

The network-wide false positive rate and statistical power are directly related. That is, as the false-positive rate increases, power, the ability of the statistical test to detect an actual release, also increases. Conversely, as the false-positive rate decreases, statistical power also decreases. Strategies to minimize the network-wide false positive rate and maximize a statistical test's power generally require careful review of the analytical data set, selection of a minimum number of representative wells and constituents

to compare, and a retesting procedure for wells when an elevated concentration is detected. A detailed discussion of these topics can be found in *Addendum to Interim Final Guidance for Statistical Analysis of Groundwater Monitoring Data at RCRA Facilities*, U.S. EPA, July 1992.. Due to the importance of these factors performing statistical analyses of groundwater data, the Discharger must also include in the Provision G.14 task f technical report a technical discussion on how it intends to (a) minimize network-wide false positive rate to less than five percent, and (b) maximize statistical power. As it continues to monitor groundwater pursuant to this program, the Discharger shall report when it deviates from the proposed Data Analysis Methods.

After one full year of groundwater monitoring, the Discharger shall analyze monitoring data from background well(s) to compute background water quality values for each monitored constituent and to perform an initial assessment of whether there is evidence of an impact from the discharge. To complete this task, the Discharger shall follow its proposed Data Analysis Methods described in the technical report required by Provision G.14 task f. Reports thereafter shall be submitted quarterly by the **1st day of the second month** after the prescribed sample collection and shall include the same analysis. The Discharger shall characterize groundwater quality using the proposed Data Analysis Method on the following constituents:

Groundwater Constituents to Evaluate Using Data Analysis Method

Alkalinity (as CaCO ₃)	Nitrate (as N)
Ammonia (as N)	Phosphate
Bicarbonate (as CaCO ₃)	Potassium
Boron	Sodium
Calcium	Sulfate
Chloride	TDS
Hardness (as CaCO ₃)	TKN
Iron	Total Nitrogen
Magnesium	TOC

WATER SUPPLY MONITORING

The supply water for City of Exeter shall be monitored as follows:

<u>Constituent</u>	<u>Units</u>	<u>Measurement</u>	<u>Frequency</u>
EC ³	µmhos/cm	Grab	Quarterly ^{2, 3}

¹ EC shall be reported as a flow-weighted average from all supply wells. Include copies of supporting calculations with monitoring reports.

² January, April, July and October

³ May be reduced to annually (October) after two years of quarterly sampling

Following two years of sampling in the manner specified, the Discharger may, following written approval by the Executive officer, establish a sampling station where a representative sample of the City's water supply can be obtained.

SLUDGE MONITORING

To ensure that discharges to the WWTF are not degrading sludge quality, the Discharger shall collect a composite sample of sludge at least annually in accordance with EPA's *POTW SLUDGE SAMPLING AND ANALYSIS GUIDANCE DOCUMENT, AUGUST 1989*, and test for metals:

Arsenic	Copper	Nickel
Cadmium	Lead	Selenium
Molybdenum	Mercury	Zinc

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling, application and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report. Prior to any disposal or land application of sewage sludge, or removal of sewage sludge from the WWTF, the monitoring and record keeping requirements of 40 CFR 503 shall be met.

REPORTING

The Discharger shall report monitoring data and information as required in this MRP and as required in the Standard Provisions and Reporting Requirements. All reports submitted in response to this MRP shall comply with the signatory requirements in Standard Provisions, General Reporting Requirements B.3. Daily, twice weekly, weekly, twice monthly, and monthly monitoring data shall be reported in monthly monitoring reports. Monthly monitoring reports shall be submitted to the Regional Board by the **1st day of second month following sampling**. Quarterly monitoring reports shall be submitted by **1st day of second month after the calendar quarter**.

Monitoring data and/or discussions submitted concerning WWTF performance must also be signed and certified by the chief plant operator. When reports contain laboratory analyses performed by the Discharger and the chief plant operator is not in the direct line of supervision of the laboratory, reports must also be signed and certified by the chief of the laboratory.

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements. If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Regional Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss any corrective actions the Discharger takes or plans to take to bring the discharge into full compliance with the waste discharge requirements.

By **1 February of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

1. The names, certificate grades, and general responsibilities of all persons in charge of wastewater treatment and disposal.

2. The names and telephone numbers of persons to contact regarding the WWTF for emergency and routine situations.
3. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4).
4. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment facility as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
5. The results of an annual evaluation conducted pursuant to Standard Provision E.4 and a figure depicting monthly average discharge flow for the past five years.
6. The most recent City of Exeter annual water supply report.
7. A summary of salinity source control accomplishments and progress towards consistent compliance with Discharge Specification B.6 (discharge EC limitation) pursuant to Provision G.13 of this Order.
8. A summary of groundwater monitoring in a format (both printed and electronic) selected in concurrence with Regional Board staff, including
 - a. Hydrographs showing the groundwater elevation in approved wells for at least the previous five years or to the extent that such data are available, whichever is fewer. The hydrographs should show groundwater elevation with respect to the elevations of the top and bottom of the screened interval and be presented at a scale of values appropriate to show trends or variations in groundwater elevation. The scale for the background plots shall be the same as that used to plot downgradient elevation data;
 - b. Graphs of the laboratory analytical data for samples taken from approved wells within at least the previous five calendar years (as data become available). Each such graph shall plot the concentration of one or more waste constituents specified below over time for a given monitoring well, at a scale appropriate to show trends or variations in water quality. The graphs shall plot each datum, rather than plotting mean values. For any given constituent, the scale for the background plots shall be the same as that used to plot downgradient data. Separate graphs shall show hydrologic equipotential gradients and equal concentration gradients for constituents below selected in concurrence with Regional Board staff.

Potential Groundwater Constituents to Evaluate

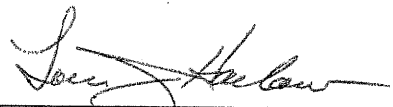
Alkalinity (as CaCO ₃)	Phosphate
Ammonia (as N)	Potassium
Boron	Sodium
Calcium	Sulfate
Chloride	TDS
Hardness (as CaCO ₃)	TKN

Potential Groundwater Constituents to Evaluate

Iron	Total Nitrogen
Magnesium	TOC
Nitrate (as N)	

- c. All monitoring analytical data obtained during the previous four quarterly reporting periods, presented in tabular form, as well as on 3.5" computer diskette.
 - d. A comprehensive discussion of the compliance record, and the result of any corrective actions taken or planned that may be needed to bring the Discharger into full compliance with the waste discharge requirements.
9. A summary of sludge monitoring, including
- a. Annual sludge production in dry tons and percent solids.
 - b. A schematic diagram showing sludge handling facilities and solids flow diagram.
 - c. A description of disposal methods, including the following information related to the disposal methods used at the WWTF. If more than one method is used, include the percentage of annual sludge production disposed of by each method.
 - i. For **landfill disposal**, include: (a) the Order numbers of WDRs that regulate the landfill(s) used, (b) the present classifications of the landfill(s) used, and (c) the names and locations of the facilities receiving sludge.
 - ii. For **land application**, include: (a) the locations of the site(s), and (b) the Order numbers of any WDRs that regulate the site(s).
 - iii. For **incineration**, include: (a) the names and location of the site(s) where sludge incineration occurs, (b) the Order numbers of WDRs that regulate the site(s), (c) the disposal method of ash, and (d) the names and locations of facilities receiving ash (if applicable).
 - iv. For **composting**, include: (a) the location of the site(s), and (b) the Order numbers of any WDRs that regulate the site(s).
10. A summary and discussion of the compliance record for the reporting period. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with this Order.

The Discharger shall implement the above monitoring program no later than 1 March 2004.


THOMAS R. PINKOS, Executive Officer

23 February 2004

(Date)